Same Squares, Different Ratios

Separating what a Punnett square produces — genotypes — from what a pathway decides afterwards: which of those genotypes end up looking alike.

Name ______________ Date ______________ Period ______________

Demonstration: andresdbp.github.io/resources/demos/same-squares/

1 Before you open anything

Answer from what you already know. The point is to have a commitment on paper that the demonstration can settle one way or the other.

PREDICTION

Two sweet pea plants heterozygous at two independently assorting genes are crossed: **CcPp × CcPp**. The offspring come out **9 purple : 7 white** — two phenotypes, not the four of a 9:3:3:1 cross. Something about this cross is different from the textbook dihybrid. Circle what you think is different, and add anything you would say instead:

the gametes each parent makes · the 16 genotypes filling the square · something that happens after the genotypes exist

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Now write the ratio you expect for the four dominance classes **C_P_ : C_pp : ccP_ : ccpp** among those 16 boxes.

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One sentence: how did you get that ratio?

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2 Read the two tallies

Open the demonstration. The square is on the left. On the right, two panels count the same 16 boxes in two different ways — that difference is the whole point.

1. In **The cross** panel, make sure the **Dihybrid 4×4** tab is selected, then press the **Complete dominance 9:3:3:1** preset.

2. In **Genotype tally — does not move**, record the number printed under **Dominance classes**, and the count on each of the four rows beneath it.
3. In **Phenotype tally — you decide this**, record the number printed under **Phenotype ratio**, and the count beside each name in the color key.

Genotype tally row	Count / 16	Phenotype name in the color key	Count / 16
R_Y_			
R_yy			
rrY_			
rryy			

a. Write the **Dominance classes** number and the **Phenotype ratio** number side by side. For this preset, what is the relationship between them?

b. At the bottom of the genotype panel is a line beginning "All 9 individual genotypes, in order". Copy the nine numbers here.

c. One of those nine numbers is 4 — the double heterozygote *RrYy*. Using the gamete row and column headings on the square, explain why *RrYy* fills four boxes while *RRYY* fills only one.

3 Six presets, one square

Press each dihybrid preset in turn, top to bottom. Fill the row before moving on — the pattern only shows up if you have all six written down.

Preset	Dominance classes	Phenotype ratio	Number of names in the color key
Complete dominance			
Complementary genes			
Dominant epistasis			
Recessive epistasis			
Duplicate recessive			
Duplicate dominant			

a. One column of your table is identical in all six rows. Which one, and what is the value?

b. Add up the counts in the color key for any one preset. Do it for a second preset. What is the total each time, and why does it have to be that number?

c. State the rule in your own words: what does changing the preset change, and what does it leave alone?

d. Go back to your prediction in part 1. Compare it with the row you recorded for **Complementary genes**. If your prediction does not match, say specifically which assumption it rested on.

> Read the **Where the model stops** panel before you generalize any of this. Every ratio on the page assumes the two genes sit on different chromosomes, that all 16 gamete pairings are equally likely, and that the offspring have been counted in large numbers. A litter of eight puppies will almost never land on 9 : 3 : 4.

4 Working backwards from the ratio

A question rarely hands you the mechanism. It hands you the numbers and asks what must have happened.

a. A dihybrid cross gives **12 : 3 : 1**. The only counts available are 9, 3, 3 and 1. Which dominance classes must have been grouped together to make the 12? Write the grouping before you check.

Now press **Dominant epistasis** and compare against the color key. Were you right?

1. Press **Reset grouping**, then **Complete dominance** again.
2. Click the bottom-right box — the *rryy* one — three times, until it is the same color as the three *rrY_* boxes. (Every box sharing a genotype moves together, which is why one click here moves exactly one box.)
3. Record both panels below.

State	Dominance classes	Phenotype ratio
Complete dominance, untouched		
After three clicks on *rryy*		

b. You have just built recessive epistasis by hand without changing a single gamete. Which of the two numbers moved, and what physically corresponds to the clicking you did?

c. Two of the six presets print exactly the same phenotype ratio. Name them. Then state what a 9 : 7 result on its own does *not* tell you, and name one kind of evidence that would settle it.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself — and where you were wrong, write down what the square would have told you.

1. A cross *AaBb* × *AaBb* gives **15 : 1**. State which dominance classes make up the 15 and which makes the 1, with counts out of 16. Then find what fraction of the 15 offspring are homozygous at *both* genes.

2. In summer squash, *WwYy* × *WwYy* gives 12 white : 3 yellow : 1 green. Of the 12 white-fruited offspring, what fraction is expected to have the genotype *WwYy*? Give it as a fraction in lowest terms and show where each number came from.

3. Labrador coat color gives 9 black : 3 chocolate : 4 yellow. Explain — no arithmetic required — why the group of 4 yellow dogs cannot all share one genotype, and describe one observation or cross that would show the yellow dogs are not genetically identical at both loci.

4. A white-eyed female fly (X^aX^a) is crossed with a red-eyed male (X^AY). Give the genotypes and eye colors of the daughters and of the sons. Then do the reciprocal cross — red-eyed female with two normal alleles (X^AX^A) with a white-eyed male (X^aY) — and explain, using the word **hemizygous**, why the two crosses give different results when a gene on an ordinary chromosome would give the same result both ways.

> Check problem 4 on the **Sex-linked** tab using the **Mother** and **Father** buttons, then the **Swap parents (reciprocal cross)** button. Read the **Daughters** and **Sons** rows of the table, not just the overall ratio — a cross where every son differs from every daughter can still print a tidy 1 : 1.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet used.

- On the **Monohybrid 2×2** tab, incomplete dominance and codominance both print 1 : 2 : 1. Read the note under each preset and decide what you would have to look at — on the organism, not on the square — to tell the two apart.
- On the **Sex-linked** tab, set the mother to X^AX^a. The **Swap parents** button switches off. Read the hint underneath and work out why a carrier mother has no male counterpart.
- Clicking moves every box with the same genotype at once, so you can split the nine genotypes into groups no preset offers. Try to build four groups of exactly 4 out of the 16 boxes. Which genotypes did you have to put together, and could any real pathway group them that way?